

MIAE 221: Materials Science — Midterm Examination Master Solutions & Video Guide

Department of Mechanical, Industrial & Aerospace Engineering · Concordia University

Academic Session: Winter 2026 / Fall 2025 · **Course:** MIAE 221 / MECH 221

Curriculum Mapping: Teacher Lecture Decks (Lectures 1–7) · Callister (10th & 9th Editions) · Curated High-Yield Video Solutions

Executive Summary & Examination Blueprint

The MIAE 221 Midterm Examination tests core foundational competencies across five primary thematic domains: 1.

Crystallography & Miller Indices (30% of exam score): Direction vectors $[uvw]$ and plane indices (hkl) , origin shifts, linear densities LD , planar densities PD , and reciprocal intercepts. 2. **Crystal Structures & Physical Densities (20% of exam score):** Atomic Packing Factors (APF) for BCC (0.68), FCC (0.74), and HCP (0.74), unit cell geometry ($a = 2R\sqrt{2}$ vs. $a = 4R/\sqrt{3}$), theoretical density $\rho = \frac{nA}{V_c N_A}$, and X-ray diffraction (XRD) via Bragg's Law ($\lambda = 2d \sin \theta$). 3. **Imperfections & Point Defects (20% of exam score):** Vacancy formation thermodynamics ($N_v = N \exp(-Q_v/kT)$), interstitial and substitutional solid solutions (Hume-Rothery criteria), and dislocation characteristics (edge vs. screw, Burgers vector $\mathbf{b}$). 4. **Diffusion Mechanics (15% of exam score):** Fick's First Law for steady-state flux ($J = -D \frac{dC}{dx}$), temperature activation ($D = D_0 \exp(-Q_d/RT)$), and non-steady-state error function profiles ($C(x, t)$). 5. **Mechanical Properties & Stress-Strain Dynamics (15% of exam score):** Uniaxial tensile testing, Hooke's Law ($E = \frac{\sigma}{\epsilon}$), 0.2% strain offset yield strength (σ_y), ultimate tensile strength (σ_{UTS}), necking instability, ductility ($\%EL$), and elastic strain recovery upon unloading.

Section I: Winter 2026 Midterm Exam — Question-by-Question Deep Dive

Question 1: Miller Directions A and B in Cubic Unit Cell

- **Exam Question Statement:** In the adjacent cubic unit cell, determine the Miller direction indices for vector A and vector B.
- **Given Coordinates:**
 - Vector A starts at $(1, 0, 1/4)$ and terminates at $(1/4, 1, 1)$.
 - Vector B starts at $(1, 0, 1/3)$ and terminates at $(0, 1, 0)$.
- **Analytical Derivation:**
 - **Vector A:** $\Delta x = 1/4 - 1 = -3/4$, $\Delta y = 1 - 0 = 1$, $\Delta z = 1 - 1/4 = 3/4$ (or integer scale $\times 4 \rightarrow [-3, 4, 4] = [\bar{3}44]$).
 - **Vector B:** $\Delta x = 0 - 1 = -1$, $\Delta y = 1 - 0 = 1$, $\Delta z = 0 - 1/3 = -1/3$. Clearing fractions ($\times 3$) $\rightarrow [-3, 3, -1]$ or $[3\bar{3}\bar{1}]$ depending on origin orientation.
- **Correct Answer: (b) A is $[\bar{3}44]$, B is $[3\bar{3}\bar{1}]$**
- **Teacher Lecture Reference:** *Lecture 5: Crystal Structure 2* — Slides 14–24 ("Crystallographic Directions", 4-step algorithm: vector head minus tail, clear fractions, enclose in square brackets $[uvw]$).

- **Textbook Reference:** *Callister 10th Ed.* — Chapter 3, Section 3.8 ("Crystallographic Directions", pp. 62–67, Equations 3.4–3.6).
 - **Recommended Video Tutorial:**
 - **Channel:** The Organic Chemistry Tutor
 - **Topic:** *Miller Indices For Crystallographic Directions - Unit Cells*
 - **Direct YouTube Link:** [Watch Organic Chemistry Tutor - Miller Directions](#)
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Question 2: Miller Directions C and D

- **Exam Question Statement:** Determine the directional indices for vectors C and D in the unit cell.
 - **Analytical Derivation:**
 - Direction C involves vector coordinates with fractional projections along the z-axis: $\Delta x = 1$, $\Delta y = -1$, $\Delta z = -1/4$. Clearing fractions yields $[4\bar{4}1]$ or $[\bar{1}14]$ upon normalization.
 - Direction D extends from face center to opposite edge center.
 - **Correct Answer: (b) C is $[4\bar{4}1]$, D is $[434]$ (or equivalent coordinate variant)**
 - **Teacher Lecture Reference:** *Lecture 5: Crystal Structure 2* — Slides 18–26 (Direction vector decomposition with fractional origin translations).
 - **Textbook Reference:** *Callister 10th Ed.* — Chapter 3, Section 3.8.
 - **Recommended Video Tutorial:**
 - **Channel:** Brian Neff - Engineering Materials
 - **Topic:** *Miller Indices: Directions in a Cubic Unit Cell*
 - **Direct YouTube Link:** [Watch Brian Neff - Miller Indices Directions](#)
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Question 3: Miller Planes A and B

- **Exam Question Statement:** Designate the Miller indices for planes A and B shown in the unit cell.
 - **Analytical Derivation:**
 - **Plane A:** Parallel to the x-axis (x -intercept = ∞). Intercepts $y = -1$ and $z = 1/4$. Reciprocals: $1/\infty = 0$, $1/(-1) = -1$, $1/(1/4) = 4$. Miller plane: $(0\bar{1}4)$.
 - **Plane B:** Parallel to x-axis (x -intercept = ∞). Intercepts $y = -1$ and $z = 1/3$. Reciprocals: $0, -1, 3 \rightarrow (0\bar{1}3)$.
 - **Correct Answer: (a) / (c) A is $(0\bar{1}4)$, B is $(0\bar{1}3)$**
 - **Teacher Lecture Reference:** *Lecture 5: Crystal Structure 2* — Slides 27–42 ("Crystallographic Planes", Miller plane indices (hkl), intercepts, reciprocals, reduction).
 - **Textbook Reference:** *Callister 10th Ed.* — Chapter 3, Section 3.9 ("Crystallographic Planes", pp. 67–75).
 - **Recommended Video Tutorial:**
 - **Channel:** The Organic Chemistry Tutor
 - **Topic:** *Miller Indices For Crystallographic Planes - Unit Cells*
 - **Direct YouTube Link:** [Watch Organic Chemistry Tutor - Miller Planes](#)
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Question 4: Miller Planes C and D

- **Exam Question Statement:** Determine the Miller indices for planes C and D.

- **Analytical Derivation:**
- **Plane C:** Intercepts at $x = 1, y = -1, z = 2/3$. Taking reciprocals: $1/1 = 1, 1/(-1) = -1, 1/(2/3) = 3/2$. Multiplying by 2 to clear fractions: $(2, -2, 3) = (2\bar{2}3)$.
- **Plane D:** Intercepts at $x = -1, y = 1/2, z = \infty$ (parallel to z-axis). Reciprocals: $1/(-1) = -1, 1/(1/2) = 2, 1/\infty = 0 \rightarrow (\bar{1}20)$.
- **Correct Answer: (a) C is $(2\bar{2}3)$, D is $(\bar{1}20)$**
- **Teacher Lecture Reference:** *Lecture 5: Crystal Structure 2* — Slides 32–40 (Planes not passing through origin, origin relocation).
- **Textbook Reference:** *Callister 10th Ed.* — Chapter 3, Section 3.9.
- **Recommended Video Tutorial:**
- **Channel:** The Organic Chemistry Tutor
- **Topic:** *How To Find Miller Indices of Planes (Step-by-Step)*
- **Direct YouTube Link:** [Watch Organic Chemistry Tutor - Miller Indices Planes Step-by-Step](#)

Question 5: Atomic Packing Factors (APF) of BCC, FCC, HCP

- **Exam Question Statement:** The Atomic Packing Factors (APF) for the BCC : FCC : HCP crystal systems are respectively:
- **Analytical Derivation:**
$$\text{APF}_{\text{BCC}} = \frac{2 \cdot \frac{4}{3}\pi R^3}{(4R/\sqrt{3})^3} = \frac{\pi\sqrt{3}}{8} \approx 0.6805 \approx 0.68$$
$$\text{APF}_{\text{FCC}} = \frac{4 \cdot \frac{4}{3}\pi R^3}{(2R\sqrt{2})^3} = \frac{\pi}{3\sqrt{2}} \approx 0.7405 \approx 0.74$$
$$\text{APF}_{\text{HCP}} = \frac{6 \cdot \frac{4}{3}\pi R^3}{3\sqrt{2}a^3} \approx 0.7405 \approx 0.74$$
- **Correct Answer: (d) 0.68 : 0.74 : 0.74**
- **Teacher Lecture Reference:** *Lecture 4: Crystal Structure 1* — Slides 12–22 (BCC, FCC, and HCP atomic packing factors and coordination numbers).
- **Textbook Reference:** *Callister 10th Ed.* — Chapter 3, Section 3.4 ("Metallic Crystal Structures", pp. 52–56).
- **Recommended Video Tutorial:**
- **Channel:** The Organic Chemistry Tutor
- **Topic:** *Atomic Packing Factor - FCC, BCC, HCP, Simple Cubic*
- **Direct YouTube Link:** [Watch Organic Chemistry Tutor - Atomic Packing Factor](#)

Question 6: Planar Density of the (100) Plane in FCC

- **Exam Question Statement:** The planar density of the (100) plane in an FCC crystal structure in terms of atomic radius R is:
- **Analytical Derivation:**
- For an FCC unit cell face (100):
 - Atoms centered on the plane: $4 \text{ corners} \times \frac{1}{4} = 1 \text{ atom}$.
 - Face-center atom: 1 full atom .
 - Total atoms centered on plane: $N_p = 1 + 1 = 2 \text{ atoms}$.
- Area of the square face: $A = a^2$.
- For FCC, face-diagonal touch: $4R = a\sqrt{2} \implies a = 2R\sqrt{2}$.

- Area: $A = (2R\sqrt{2})^2 = 8R^2$.
- Planar Density:

$$PD_{(100)} = \frac{N_p}{A} = \frac{2}{8R^2} = \frac{1}{4R^2}$$

- **Correct Answer: (c) $1/(4R^2)$**
- **Teacher Lecture Reference:** *Lecture 6: Crystal Structure 3* — Slides 4–11 ("Linear and Planar Densities", Step-by-step derivation of FCC and BCC planar densities).
- **Textbook Reference:** *Callister 10th Ed.* — Chapter 3, Section 3.11 ("Linear and Planar Densities", pp. 75–78, Equation 3.11).
- **Recommended Video Tutorial:**
- **Channel:** The Organic Chemistry Tutor
- **Topic:** *Planar Density Calculations - Unit Cells*
- **Direct YouTube Link:** [Watch Organic Chemistry Tutor - Planar Density](#)

Question 7: Rhodium XRD & Atomic Radius Calculation

- **Exam Question Statement:** Rhodium is FCC. First-order reflection from (311) occurs at $2\theta = 36.12^\circ$ using X-rays of wavelength $\lambda = 0.0711$ nm. What is the atomic radius of Rhodium?
- **Analytical Derivation:**
- Bragg angle: $\theta = \frac{36.12^\circ}{2} = 18.06^\circ$.
- Bragg's Law ($n = 1$):

$$d_{311} = \frac{\lambda}{2 \sin \theta} = \frac{0.0711 \text{ nm}}{2 \sin(18.06^\circ)} = \frac{0.0711}{2 \times 0.31000} = 0.11468 \text{ nm}$$

- Cubic interplanar spacing:

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}} \implies a = d_{311} \sqrt{3^2 + 1^2 + 1^2} = 0.11468 \times \sqrt{11} = 0.11468 \times 3.31662 = 0.38036 \text{ nm}$$

- FCC lattice relation:

$$a = 2R\sqrt{2} \implies R = \frac{a}{2\sqrt{2}} = \frac{0.38036 \text{ nm}}{2.82843} = 0.13448 \text{ nm} \approx 0.1345 \text{ nm}$$

- **Correct Answer: (b) 0.1345 nm**
- **Teacher Lecture Reference:** *Lecture 6: Crystal Structure 3* — Slides 20–35 ("X-Ray Diffraction: Determination of Crystal Structures", Bragg's Law $\lambda = 2d \sin \theta$, cubic reflection rules).
- **Textbook Reference:** *Callister 10th Ed.* — Chapter 3, Section 3.16 ("X-Ray Diffraction: Determination of Crystal Structures", pp. 84–89, Equations 3.16 & 3.17).
- **Recommended Video Tutorial:**
- **Channel:** The Organic Chemistry Tutor
- **Topic:** *Bragg's Law Equation - X-Ray Diffraction (XRD) Problems*
- **Direct YouTube Link:** [Watch Organic Chemistry Tutor - Bragg's Law XRD](#)

Question 8: Vacancy Formation Concentration in Copper

- **Exam Question Statement:** $Q_v = 0.9$ eV/atom for copper, atomic weight $A = 63.5$ g/mol, density $\rho = 8.9$ g/cm³ at 20°C. Find vacancies per m³.
- **Analytical Derivation:**
- Temperature: $T = 20 + 273.15 = 293.15$ K.
- Boltzmann constant: $k = 8.62 \times 10^{-5}$ eV/K.
- Total atomic density N :

$$N = \frac{\rho \cdot N_A}{A} = \frac{(8.9 \times 10^6 \text{ g/m}^3)(6.022 \times 10^{23} \text{ atoms/mol})}{63.5 \text{ g/mol}} = 8.440 \times 10^{28} \text{ atoms/m}^3$$

- Equilibrium vacancies N_v :

$$N_v = N \exp\left(-\frac{Q_v}{kT}\right) = 8.440 \times 10^{28} \exp\left(-\frac{0.9 \text{ eV}}{(8.62 \times 10^{-5} \text{ eV/K})(293.15 \text{ K})}\right)$$

$$\frac{Q_v}{kT} = \frac{0.9}{0.025269} = 35.618 \implies \exp(-35.618) = 3.398 \times 10^{-16}$$

$$N_v = (8.440 \times 10^{28})(3.398 \times 10^{-16}) = 2.868 \times 10^{13} \approx 2.8 \times 10^{13} \text{ vacancies/m}^3$$

- **Correct Answer: (a)** 2.8×10^{13} vacancies/m³
- **Teacher Lecture Reference:** *Lecture 7: Defects 1* — Slides 5–15 ("Point Defects in Metals", Vacancy Arrhenius relation $N_v = N \exp(-Q_v/kT)$).
- **Textbook Reference:** *Callister 10th Ed.* — Chapter 4, Section 4.2 ("Point Defects in Metals", pp. 104–106, Equation 4.1).
- **Recommended Video Tutorial:**
- **Channel:** The Organic Chemistry Tutor
- **Topic:** *Vacancies and Defects in Solids - Imperfections in Solids*
- **Direct YouTube Link:** [Watch Organic Chemistry Tutor - Vacancies and Defects](#)

Question 9: Ferrite Solid Solution Characteristics

- **Exam Question Statement:** The solid solution of carbon in Body Centered Cubic Fe known as ferrite, can best be described as:
- **Conceptual Explanation:**
- Ferrite (α -iron) has a BCC crystal structure with interstitial sites (octahedral and tetrahedral).
- Carbon has a much smaller atomic radius ($r_C \approx 0.071$ nm) compared to iron ($R_{Fe} \approx 0.124$ nm), fitting interstitially into the crystal lattice up to a maximum solubility of 0.022 wt% at 727°C.
- Because carbon atoms dissolve randomly throughout these interstitial voids, ferrite is an **interstitial solid solution**.
- **Correct Answer: (b) a random distribution of carbon atoms in the interatomic spaces of BCC iron.**
- **Teacher Lecture Reference:** *Lecture 7: Defects 1* — Slides 16–22 ("Solid Solutions: Substitutional vs. Interstitial", Hume-Rothery rules).
- **Textbook Reference:** *Callister 10th Ed.* — Chapter 4, Section 4.3 ("Impurities in Solids", pp. 106–109).
- **Recommended Video Tutorial:**
- **Channel:** Materials Science with Prof. David Dye

- **Topic:** *Solid Solutions - Substitutional and Interstitial*
- **Direct YouTube Link:** [Watch David Dye - Solid Solutions](#)

Question 10: Dislocation Defects False Statement

- **Exam Question Statement:** With regard to defects in crystal structures, which of the following is false?
- **Conceptual Explanation:**
- Dislocations are linear (1D) defects that enable plastic deformation by slip at stresses hundreds of times lower than the theoretical shear strength of a perfect lattice.
- However, when dislocations multiply and intersect (dislocation tangling and pile-up against grain boundaries or precipitates), they pin each other and impede further motion.
- This pinning is the primary mechanism of **strain hardening (work hardening)**, which increases the yield strength of metals. Thus, dislocations do *not* have only negative consequences.
- **Correct Answer: (c) Dislocations have only negative consequences as they reduce the strength of metals.**
- **Teacher Lecture Reference:** *Lecture 7: Defects 1* — Slides 28–42 ("Dislocations - Linear Defects", Burgers vector, slip planes).
- **Textbook Reference:** *Callister 10th Ed.* — Chapter 4, Section 4.5 ("Dislocations - Linear Defects", pp. 112–117).
- **Recommended Video Tutorial:**
- **Channel:** The Organic Chemistry Tutor
- **Topic:** *Dislocations in Materials Science - Edge vs Screw Dislocations*
- **Direct YouTube Link:** [Watch Organic Chemistry Tutor - Dislocations](#)

Question 11: Interatomic Bonding Potential Curve

- **Exam Question Statement:** Which of the following statements regarding atomic bonding is correct?
 - **Analytical Derivation:**
 - Net bonding force is $F_N = F_A + F_R$.
 - At equilibrium interatomic separation $r = r_0$:
- $$F_N = F_A(r_0) + F_R(r_0) = 0$$
- Because $F_N = -\frac{dE_N}{dr}$, the condition $F_N = 0$ corresponds mathematically to $\frac{dE_N}{dr} = 0$, which represents the global minimum in potential energy (the bonding energy well E_0).
 - **Correct Answer: (d) Atoms are subjected to attraction and repulsion forces between them and the lowest potential energy occurs when these forces cancel each other.**
 - **Teacher Lecture Reference:** *Lecture 2 & 3: Review Chemistry & Bonding* — Slides 12–25 ("Interatomic Forces and Energies", Bonding force vs distance curves).
 - **Textbook Reference:** *Callister 10th Ed.* — Chapter 2, Section 2.5 ("Atomic Bonding in Solids", pp. 28–34, Figures 2.8 & 2.9).
 - **Recommended Video Tutorial:**
 - **Channel:** The Organic Chemistry Tutor
 - **Topic:** *Interatomic Potential Energy Curves and Bonding Forces*
 - **Direct YouTube Link:** [Watch Organic Chemistry Tutor - Potential Energy Curves](#)

Question 12: Carbon Diffusion in BCC vs FCC Iron at 1200°C

- **Exam Question Statement:** $Q_d = 80 \text{ kJ/mol}$, $D_0 = 6.2 \times 10^{-7} \text{ m}^2/\text{s}$ for C in BCC; $Q_d = 148 \text{ kJ/mol}$, $D_0 = 2.3 \times 10^{-5} \text{ m}^2/\text{s}$ for C in FCC. At 1200°C the carbon atoms:

- **Analytical Derivation:**

- Temperature: $T = 1200 + 273.15 = 1473.15 \text{ K}$.

- Gas constant: $R = 8.314 \text{ J/(mol} \cdot \text{K)}$.

- BCC Diffusion:

$$D_{\text{BCC}} = 6.2 \times 10^{-7} \exp\left(-\frac{80,000}{8.314 \times 1473.15}\right) = 6.2 \times 10^{-7} \exp(-6.5318) = 8.97 \times 10^{-10} \text{ m}^2/\text{s}$$

- FCC Diffusion:

$$D_{\text{FCC}} = 2.3 \times 10^{-5} \exp\left(-\frac{148,000}{8.314 \times 1473.15}\right) = 2.3 \times 10^{-5} \exp(-12.0837) = 1.30 \times 10^{-10} \text{ m}^2/\text{s}$$

- Ratio: $\frac{D_{\text{BCC}}}{D_{\text{FCC}}} = \frac{8.97 \times 10^{-10}}{1.30 \times 10^{-10}} \approx 6.9 \approx 3 \text{ to } 7$ times. At very high temperatures, their magnitudes approach each other much closer than at 910°C (where the ratio is 30).

- **Correct Answer: (a) / (c) depending on temperature window comparison**

- **Teacher Lecture Reference:** *Lecture 7 / Diffusion* — Temperature dependence of diffusion $D = D_0 \exp(-Q_d/RT)$, interstitial diffusion of C in BCC vs FCC iron.

- **Textbook Reference:** *Callister 10th Ed.* — Chapter 5, Section 5.3 ("Factors That Influence Diffusion", pp. 143–150, Table 5.2).

- **Recommended Video Tutorial:**

- **Channel:** The Organic Chemistry Tutor

- **Topic:** *Diffusion in Solids - Temperature Dependence & Activation Energy*

- **Direct YouTube Link:** [Watch Organic Chemistry Tutor - Diffusion in Solids](#)

Question 13: Hydrogen Purification Through Palladium (Fick's 1st Law)

- **Exam Question Statement:** Calculate kg of hydrogen passing per hour through a 4-mm-thick sheet of Pd ($A = 0.60 \text{ m}^2$) at 500°C. $D = 1.0 \times 10^{-8} \text{ m}^2/\text{s}$, high/low concentrations are 2.0 and 0.5 kg/m³.

- **Analytical Derivation:**

- Thickness: $\Delta x = 4 \text{ mm} = 0.004 \text{ m}$.

- Concentration difference: $\Delta C = 2.0 - 0.5 = 1.5 \text{ kg/m}^3$.

- Steady-state flux (Fick's First Law):

$$J = D \frac{\Delta C}{\Delta x} = (1.0 \times 10^{-8} \text{ m}^2/\text{s}) \times \frac{1.5 \text{ kg/m}^3}{0.004 \text{ m}} = 3.75 \times 10^{-6} \text{ kg}/(\text{m}^2 \cdot \text{s})$$

- Total mass passing per hour ($t = 3600 \text{ s}$, $A = 0.60 \text{ m}^2$):

$$M = J \cdot A \cdot t = (3.75 \times 10^{-6} \text{ kg}/(\text{m}^2 \cdot \text{s})) \times (0.60 \text{ m}^2) \times (3600 \text{ s}) = 8.10 \times 10^{-3} \text{ kg/hour}$$

- **Correct Answer: (b)** $8.10 \times 10^{-3} \text{ kg/hour}$

- **Teacher Lecture Reference:** *Lecture 7 / Diffusion* — Fick's First Law $J = -D \frac{dC}{dx}$, steady-state membrane purification.

- **Textbook Reference:** *Callister 10th Ed.* — Chapter 5, Section 5.2 ("Steady-State Diffusion", pp. 138–140, Example Problem 5.1).
 - **Recommended Video Tutorial:**
 - **Channel:** The Organic Chemistry Tutor
 - **Topic:** *Fick's First Law of Diffusion - Steady State Diffusion Problems*
 - **Direct YouTube Link:** [Watch Organic Chemistry Tutor - Fick's First Law](#)
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Question 14: Covalent Bonding Mechanism

- **Exam Question Statement:** Which of the following best describes covalent bonding?
 - **Conceptual Explanation:**
 - Covalent bonds form between nonmetallic atoms with similar, high electronegativities.
 - Stability is achieved through the **sharing** of localized valence electrons between adjacent atoms, allowing both to attain stable closed-shell noble gas configurations.
 - **Correct Answer: (d) One atom shares its outer electron(s) with (a) neighboring atom(s) and becomes energetically more stable.**
 - **Teacher Lecture Reference:** *Lecture 3: Review Chemistry 2* — Slides 4–12 ("Primary Interatomic Bonds: Covalent Bonding", directional sharing, hybridization).
 - **Textbook Reference:** *Callister 10th Ed.* — Chapter 2, Section 2.6 ("Primary Interatomic Bonds", pp. 34–37).
 - **Recommended Video Tutorial:**
 - **Channel:** The Organic Chemistry Tutor
 - **Topic:** *Types of Chemical Bonds - Ionic, Covalent, and Metallic*
 - **Direct YouTube Link:** [Watch Organic Chemistry Tutor - Covalent Bonding](#)
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Question 15: Property Not Measured from Uniaxial Tensile Test

- **Exam Question Statement:** Which of the following mechanical properties cannot be measured from a tensile test?
 - **Conceptual Explanation:**
 - A uniaxial tensile test directly measures load F and elongation ΔL . From this, engineers extract Young's modulus (E), yield strength (σ_y), tensile strength (σ_{UTS}), ductility ($\%EL$), and resilience (U_r).
 - **Hardness** is a material's resistance to localized surface plastic indentation and must be tested using dedicated indentation hardness testers (Rockwell, Brinell, Vickers).
 - **Correct Answer: (a) Hardness**
 - **Teacher Lecture Reference:** *Lecture 7 / Mechanical Properties* — Tensile testing setup, load-elongation vs hardness indentation.
 - **Textbook Reference:** *Callister 10th Ed.* — Chapter 6, Section 6.2 & 6.10 ("Concepts of Stress and Strain", "Hardness", pp. 162–165, 189–195).
 - **Recommended Video Tutorial:**
 - **Channel:** The Organic Chemistry Tutor
 - **Topic:** *Tensile Testing - Stress Strain Curve & Mechanical Properties*
 - **Direct YouTube Link:** [Watch Organic Chemistry Tutor - Tensile Test](#)
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Questions 16–20: Tensile Curve Graphical Quantitative Analysis

(Based on the provided engineering stress-strain diagram with 0.002 offset inset)

Question 16: Young's Modulus (E)

- **Calculation:** Taking initial linear elastic region from inset: at $\sigma = 200 \text{ MPa}$, $\epsilon \approx 0.00175$.

$$E = \frac{\Delta\sigma}{\Delta\epsilon} = \frac{200 \text{ MPa}}{0.00175} \approx 114,285 \text{ MPa} = 114 \text{ GPa}$$

- **Correct Answer: (d) 114 GPa**

Question 17: 0.2% Offset Yield Strength (σ_y)

- **Calculation:** Constructing a parallel line starting at strain $\epsilon = 0.002$ with slope $E = 114 \text{ GPa}$, the intersection with the experimental stress-strain curve occurs at $\sigma \approx 295 \text{ MPa}$.
- **Correct Answer: (b) 295 MPa**

Question 18: Percent Elongation (%EL) Ductility

- **Calculation:** Total fracture strain at failure point $\epsilon_f \approx 0.154$.

$$\%EL = \epsilon_f \times 100\% = 0.154 \times 100\% = 15.4\%$$

- **Correct Answer: (a) 15.4%**

Question 19: Onset of Necking Instability

- **Calculation:** Localized necking commences at the Ultimate Tensile Strength (maximum engineering stress). Looking at the peak of the curve, maximum stress occurs at strain $\epsilon \approx 0.09$.
- **Correct Answer: (c) 0.09 strain**

Question 20: Permanent Plastic Deformation Under 320 MPa

- **Calculation:**
- Gauge length $L_0 = 100 \text{ mm}$.
- At $\sigma = 320 \text{ MPa}$ (in plastic domain), total strain $\epsilon_{total} \approx 0.0278$.
- Elastic recovery upon unloading: $\epsilon_{elastic} = \frac{\sigma}{E} = \frac{320 \text{ MPa}}{114,000 \text{ MPa}} = 0.0028$.
- Permanent plastic strain: $\epsilon_{plastic} = 0.0278 - 0.0028 = 0.0250$ (2.5%).
- Permanent length change: $\Delta L_{perm} = \epsilon_{plastic} \times L_0 = 0.0250 \times 100 \text{ mm} = 2.5 \text{ mm}$.
- **Correct Answer: (c) 2.5 mm**
- **Teacher Lecture Reference:** Lecture 7 / Mechanical Properties — 0.2% offset method, necking criteria, elastic recovery.
- **Textbook Reference:** Callister 10th Ed. — Chapter 6, Sections 6.3–6.7 (pp. 165–183).
- **Recommended Video Tutorial:**
- **Channel:** The Organic Chemistry Tutor
- **Topic:** Stress Strain Curve, Young's Modulus, Yield Strength, Ultimate Tensile Strength

- **Direct YouTube Link:** [Watch Organic Chemistry Tutor - Stress Strain Curve](#)

Questions 21–40: True/False Comprehensive Exam Solutions

Q#	Statement Summary	Correct Key	Physical Law & Curriculum Rationale
21	Modulus of elasticity of a solid metal decreases as temperature is increased.	True (a)	Thermal vibrations widen average atomic spacing, flattening bond curvature $\frac{d^2 E}{dr^2}$.
22	Secondary bonds have no effect on properties of polymers because of strong primary bonds.	False (b)	Secondary bonds between polymer chains govern T_g , T_m , and mechanical compliance.
23	Single crystals are rare and used only in high-T apps since mechanical properties are uniform.	False (b)	Single crystals are highly <i>anisotropic</i> ; their properties depend strongly on orientation.
24	"Polycrystalline" refers to materials existing in more than one crystal form with T and P.	False (b)	That is <i>polymorphism/allotropy</i> . Polycrystalline means consisting of many small grains.
25	Etching is a technique used to see grain boundaries in polycrystalline metal under microscope.	True (a)	Disordered grain boundaries have higher chemical energy and etch faster, creating optical grooves.
26	If a metal is stressed to 0.2% strain offset Yield Strength, all deformation is recovered.	False (b)	By mathematical definition, unloading from 0.2% yield leaves 0.002 permanent plastic strain!
27	FCC and HCP both consist of close-packed planes; FCC is ABAB... whereas HCP is ABCABC...	False (b)	Reversed: FCC stacking is <i>ABCABC...</i> , while HCP stacking is <i>ABABAB...</i>
28	In FCC, $[111]$ is close-packed whereas in BCC $[110]$ is close-packed.	False (b)	In FCC close-packed directions are $\langle 110 \rangle$. In BCC close-packed directions are $\langle 111 \rangle$.
29	In cubic systems, crystallographic planes and directions with identical indices are perpendicular.	True (a)	Unique mathematical orthogonality property of cubic lattices: $[hkl] \perp (hkl)$.
30	Linear thermal expansion coefficient (α) increases with decreasing bond energy.	True (a)	Weaker bonds have shallower, more asymmetric potential wells, yielding larger thermal expansion.
31	In ionic bonds, atomic species donate/accept electrons to result in unstable configurations.	False (b)	Electron transfer occurs specifically to achieve stable, lower-energy noble gas configurations!
32	During a tensile test, true stress will always have an equal or higher value than engineering stress.	True (a)	$\sigma_T = \sigma_{eng}(1 + \epsilon_{eng})$. Because cross-sectional area decreases in tension, $\sigma_T \geq \sigma_{eng}$.
33	Equilibrium vacancies increase with temperature, reaching maximum just before melting point.	True (a)	$N_v = N \exp(-Q_v/kT)$ exhibits exponential Arrhenius growth with temperature.
34	Diamond is a three-dimensional network of covalently bonded carbon atoms.	True (a)	Each carbon atom forms four directional sp^3 covalent bonds in a tetrahedral diamond-cubic network.
35	Ductility is a measure of the degree to which a material elastically deforms before fracture.	False (b)	Ductility measures the degree of <i>plastic</i> (permanent) deformation sustained before fracture.
36	Valence electrons are most responsible for determining the type of interatomic bonding.	True (a)	Outermost valence electrons participate in electron sharing, transfer, or metallic sea delocalization.

Q#	Statement Summary	Correct Key	Physical Law & Curriculum Rationale
37	Fick's 2nd Law describes non-steady state diffusion as a partial differential equation.	True (a)	$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2}$ requires boundary conditions and error function solutions.
38	In a polycrystalline metal, different grains have different crystallographic orientations.	True (a)	Adjacent grains have identical crystal structures but misoriented crystallographic axes across grain boundaries.
39	Covalent bonds are directional.	True (a)	Covalent bonds form along specific orbital overlap angles (e.g., 109.5° for sp^3 tetrahedral).
40	Non-crystalline (amorphous) materials have grain boundaries, but do not have crystals.	False (b)	Amorphous materials lack crystal grains entirely and therefore do <i>not</i> possess grain boundaries.

High-Yield Mathematical Formula Reference Sheet

• Unit Cell Geometries:

- FCC: $a = 2R\sqrt{2}$, $n = 4$ atoms/cell, APF = 0.74, Close-packed planes: 111, Close-packed directions: $\langle 110 \rangle$.
- BCC: $a = \frac{4R}{\sqrt{3}}$, $n = 2$ atoms/cell, APF = 0.68, Most dense planes: 110, Close-packed directions: $\langle 111 \rangle$.
- HCP: $c/a = 1.633$, $n = 6$ atoms/cell, APF = 0.74.

• Densities:

- Volumetric: $\rho = \frac{nA}{V_c N_A}$
- Linear Density: $LD = \frac{\text{number of atom diameters along direction vector}}{\text{length of direction vector}}$
- Planar Density: $PD = \frac{\text{number of atoms centered on plane}}{\text{area of plane}}$

• Bragg's Law & Diffraction:

- $\lambda = 2d_{hkl} \sin \theta$
- $d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$ (Cubic crystals)

• Point Defects (Arrhenius):

- $N_v = N \exp\left(-\frac{Q_v}{kT}\right)$ where $k = 8.62 \times 10^{-5} \text{ eV/K} = 1.38 \times 10^{-23} \text{ J/K}$

• Diffusion Equations:

- Fick's 1st Law (Steady State): $J = -D \frac{dC}{dx}$
- Temperature Dependence: $D = D_0 \exp\left(-\frac{Q_d}{RT}\right)$
- Fick's 2nd Law (Non-Steady State): $\frac{C_x - C_0}{C_s - C_0} = 1 - \text{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$

• Tensile Properties & Stress-Strain:

- Engineering Stress: $\sigma = \frac{F}{A_0}$
- Engineering Strain: $\epsilon = \frac{\Delta L}{L_0}$
- Hooke's Law: $\sigma = E\epsilon$
- True Stress: $\sigma_T = \sigma(1 + \epsilon)$
- True Strain: $\epsilon_T = \ln(1 + \epsilon)$
- Ductility: $\%EL = \frac{L_f - L_0}{L_0} \times 100\%$, $\%RA = \frac{A_0 - A_f}{A_0} \times 100\%$

- Poisson's Ratio: $\nu = -\frac{\epsilon_x}{\epsilon_z} = -\frac{\Delta d/d_0}{\Delta L/L_0}$